Timestamp-Alignment Leakage in Drive-Test Handover Prediction: Quantification, Mitigation, and Blind Second-Operator Validation

Saadman Sakib, Adhnan Kalim, Evan Ashfaque, Mohammad Tawhid Kawser*

Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT), Board Bazar, Gazipur 1704, Bangladesh
Emails: saadmansakib21@iut-dhaka.edu, adhnankalim@iut-dhaka.edu, evanashfaque@iut-dhaka.edu, kawser@iut-dhaka.edu

***Corresponding author:**
Mohammad Tawhid Kawser
Department of Electrical and Electronic Engineering, Islamic University of Technology (IUT)
Board Bazar, Gazipur 1704, Bangladesh
Email: kawser@iut-dhaka.edu

Highlights

- One-hertz drive-test rows carry the post-handover cell before the handover command.
- Ignoring this inflates one-second handover-prediction performance four-fold.
- Serving-cell radio scalars, not cell identifiers, carry 94% of the leak.
- Splitting protocol reorders learner rankings, not just their scores.
- The leak replicates blind on a second operator; learner rankings do not.

Abstract

Machine learning models for cellular handover forecasting are frequently trained on one-hertz drive-test exports and report optimistic predictive scores. This paper demonstrates that an unexamined feature of tabular export aggregation, rather than the underlying cellular network or machine learning algorithm, can produce much of that apparent predictive performance. Across four drive-test measurement campaigns on a commercial Long Term Evolution (LTE) cellular network (10,260 one-hertz samples, 938 handover commands decoded directly from Radio Resource Control (RRC) signalling), the export row stamped in the second immediately preceding a handover command already reflects the post-handover serving cell in 76% of handovers, compared to only 16% one row earlier. Lagging all export features by a single row shifts the one-second Area Under the Precision-Recall Curve (AUPRC) of the benchmark model from 0.604 to 0.156, representing a 3.9-fold inflation; 94% of this reduction stems from serving-cell radio measurements whose referents invert upon handover execution. End-of-second aggregation on native-rate telemetry from an independent corpus spanning three operators reproduces the contamination (93%), whereas start-of-second alignment eliminates it (4%). A validation drive on a second cellular operator, conducted after freezing the training pipeline, confirms both findings: 67% row contamination and a 4.3-fold performance inflation. When this temporal leakage is removed and campaign holdout evaluation is enforced, gradient-boosted decision trees achieve an exploratory one-second AUPRC of 0.169 against a 6.8% prevalence baseline.

Keywords

Machine learning; Gradient-boosted decision trees; Discrete-time survival analysis; Conformal risk control; Data leakage audit; Timestamp alignment; Cellular handover forecasting; Telecommunication telemetry; Drive testing

CRediT authorship contribution statement

Saadman Sakib: Conceptualization, Methodology, Software, Investigation, Formal analysis, Writing – original draft.
Adhnan Kalim: Investigation, Data curation, Validation, Writing – review & editing.
Evan Ashfaque: Investigation, Data curation, Visualization, Writing – review & editing.
Mohammad Tawhid Kawser: Supervision, Resources, Methodology, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. Mohammad Tawhid Kawser is a co-author of an earlier public drive-test dataset cited in the paper (ref [38]), which was collected independently and is not used for any experimental claim or model evaluation in this study.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used an AI assistant (Claude, Anthropic) to help condense this work into journal form and to edit the language. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.